

CMPG 325 COMPUTER NETWORKS

PHASE 1 DESIGN DOCUMENT

Molema Citrus Estates (Rustenburg)

Industry: Agriculture

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Project ID	CMPG325-2026-037
Client ID	CLI-037
Assigned Organization	Molema Citrus Estates (Rustenburg)
Industry	Agriculture
Assigned Addressing Block	192.168.26.0/24
Assigned Networking Challenge	ACLs (traffic filtering policy) Intermediate
Design Constraint	VoIP traffic must be separated from data traffic
Change Request	CR14 — After-hours cleaning/security contractors require limited wireless access
Project Period	14 August 2026 – 16 October 2026
Milestone 1 Deadline	28 August 2026 -Client Design Review

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Executive Summary

This document outlines the first stage of the network setup **Molema Citrus Estate**, which is place where citrus fruits are grown in Rustenburg. The suggested network setup has a two-level structure. It includes special edge router and a separate core Lyr-3-switch. The suggested switches handle different networks like Administration, packhouse/Operations, Voice, Management, and Guest. A VLSM addressing plan uses the assigned 192.168.26.0/24 block in a way that saves 132 addresses for future use. The design fulfills all the client's confirmed needs, like separating Volp traffic on VLAN 40, providing isolated guest access on VLAN 50 as per CR14, and including a draft ACL policy that tackles the traffic-filtering issues assigned. This document meets all the requirements for Client design, it includes the client requirements the physical topology, the logical topology and the IP addressing plan.

1. INTRODUCTION

This document serves as the phase 1 Design document for **Molema Citrus Estate project**. It is submitted in fulfillment of milestone 1 client design review and presents the following core deliverables: a thorough analysis of the assigned client's requirements, the proposed physical and logical network topologies, and a VLSM-based IP addressing plan derived from allocated address block 192.168.26.0/24.

In alignment with the project scope and milestone objectives, this document does not encompass Packet Tracer configuration, ACL implementation, network testing, or the final technical report. These deliverables are scheduled for milestone 2 and the final submission, contingent upon the approval of the design presented herein.

1.1 This document presents a complete and self-contained network design for the client. It begins by establishing the clients background and requirements, then proceeds to describe the proposed physical and logical topology, the Ip addressing plan, and the justification for each key design decision. The document is structured to allow any reader to understand the solution without requiring additional context or explanation.

2. Client Background

2.1 Client Profile

Molema Citrus Estates is an agricultural organization based in Rustenburg, Northwest Province, operating within citrus farming industry. The client has been assigned to this project under Client ID: CLI-037 and the Project ID CMPG325-2026-037. The project brief explicitly requires that the proposed solution remain specific to this client and must not be substituted with a generic or alternative scenario.

ASSUMPTION

Based on typical South African citrus export estates, operation are assumed to include administrative/office functions, packhouse/processing, field operations, logistics, and security. This structure justifies the departmental segmentation in the proposed design

ASSUMPTION

Reliable connectivity is critical due to time-sensitive cold-chain operations, export deadlines, and the need for administrative system to communicate with staff suppliers, and logistics partner.

MISSING

Exact department names, headcounts, building layouts, and number of physical site are not specified in the brief. These are treated as designed assumptions section (5) and flagged for confirmation with the module coordinator prior to milestone 2

3. Client Requirements

3.1 Functional Requirements

Requirement	Source/Reason	Design Response	Priority
Provide appropriate connectivity for the assigned scenario	Client brief, Section 6	Router + core L3 switch + access switches per functional area	High
Use 192.168.26.0/24 as the basis of the IP addressing plan	Client brief, Section 6 and 7	VLSM addressing plan (Section 8)	High
Separate VoIP traffic from data traffic	Client brief, Section 8 (Design Constraint)	Dedicated Voice VLAN 40, distinct subnet, ACL between voice and data	High
Configure, verify and demonstrate ACLs (traffic filtering policy)	Client brief, Section 9 (Assigned challenge)	Draft ACL policy table (Section 15); implementation is Phase 2	High
Requirement	Source/Reason	Design Response	Priority
Accommodate CR14: after-hours contractors require limited wireless access	Client brief, Section 10 (Change Request)	Dedicated Guest/Contractor VLAN 50 on a separate wireless AP, isolated from internal VLANS	High
Produce a working, testable Packet Tracer implementation	Client brief, Section 6	Physical/logical design prepared now; build is Phase 2	High
Demonstrate successful connectivity and testing	Client brief, section 7	Testing plan to be executed in Phase 2 against this design	Medium
Document significant design decisions and evidence in GitHub	Client brief, Section 7 and 12	GitHub repository structure (Section 18)	Medium

3.2 Non-Functional Requirements

Requirement	Source/Reason	Design Response	Priority
Reliability	Client depends on continuous connectivity for time-sensitive packhouse/dispatch operations (assumption)	Structured hierarchical design with a dedicated core device; no single access switch carries all traffic	Medium
Security	VoIP separation constraint + guest access change request imply segmentation is a priority	VLAN segmentation, guest isolation, ACL policy at the core/edge	High
Efficient IP address utilization	Single /24 block (192.168.26.0/24) must serve the whole estate	VLSM sizing per segment instead of flat /24 or equal subnetting (Section 8)	High
Scalability	Rubric explicitly rewards futureproofing; agriculture estates typically grow seasonal/staff numbers	Reserved address space, spare VLAN IDs, modular access-layer design (Section 17)	Medium
Manageability	Single IT/network administrator is typical for an estate of this size (assumption)	Dedicated Management VLAN 10 for administrative access to network devices	Medium

4.Constrain

[CONFIRMED] Assigned client: Molema Citrus Estates (Rustenburg); Industry: Agriculture; Addressing block: 192.168.26.0/24.

[CONFIRMED] Design constraint: VoIP traffic must be separated from data traffic.

[CONFIRMED] Assigned networking challenge (CR14 label in rubric: 'assigned networking challenge / technical challenge'): ACLs (traffic filtering policy), classified Intermediate difficulty. The brief requires this to be configured, verified and demonstrated, with an explanation of what was configured, why it is appropriate, and how it was verified.

[CONFIRMED] Change Request CR14 (Section 10 of the brief): after-hours cleaning/security contractors require limited wireless access. The final solution must show how the network accommodates this change request specifically, without introducing additional scope.

[CONFIRMED] The project requires Cisco Packet Tracer for implementation and simulation, VLSM-based use of the assigned address block, and documentation of design decisions and evidence in GitHub.

[MISSING] The brief text supplied does not include a separate numbered 'CR14' technical specification beyond the wireless-access change request itself — the change request in Section 10 is treated as the full and only CR14 requirement. No additional CR14 technical detail is assumed or fabricated.

5. Confirmed / Assumed / Missing Information Summary

5.1 Confirmed (FROM THE CLIENT)

- Client: Molema Citrus Estates (Rustenburg); Industry: Agriculture; Project ID CMPG325-2026-037; Client ID CLI-037
- Addressing block: 192.168.26.0/24, to be used with VLSM
- Constraint: VoIP traffic separated from data traffic
- Assigned challenge: ACLs (traffic filtering policy), Intermediate difficulty.
- Change request CR14: after-hours cleaning/security contractors require limited wireless access.
- Milestones: 14 Aug (start), 28 Aug (Milestone 1), 2 Oct (Milestone 2), 16 Oct (final submission).

5.2 Reasonable design assumption

- Estate operations are organized into an Administration/Office function and a Packhouse/Operations function, plus a small IT/network management function — typical of a citrus export estate.
- Approximate host counts per segment (used only to size subnets, not presented as client-confirmed figures): Administration $\approx$ 20 devices, Packhouse/Operations $\approx$ 30 devices, VoIP handsets $\approx$ 15, Guest/contractor Wi-Fi $\approx$ 10 concurrent devices, Management $\approx$ 5 devices.
- A single physical site (the estate) is assumed, connected to the Internet via a single ISP link, with buildings/areas connected internally rather than over a WAN.
- Guest/contractor wireless access should be Internet-only and denied access to all internal VLANs, consistent with 'limited wireless access' in CR14.

5.3 Missing Information Requiring Confirmation

- Exact department names, headcounts and organizational structure at Molema Citrus Estates.
- Physical building layout/number of structures on the estate and distances between them (affects cabling choices).
- Whether any dynamic routing, redundancy (WAN failover) or specific service-level requirement is expected beyond what is stated in the brief.
- Exact ACL rule set expected beyond 'traffic filtering policy' will be finalized with the lecturer/module guidance before Phase 2 configuration.

6. Network Design Objectives

Design Objectives	Linked Client Requirement
Provide reliable connectivity across administration, operations and management functions	Appropriate connectivity for assigned scenario
Separate voice (VoIP) traffic at Layer 2 and Layer 3	Design constraints: VoIP separated from data
Use 192.168.26.0/24 efficiently through VLSM rather than flat subnetting	Assigned addressing block must be the basis of the IP plan
Filter traffic between network segments using ACLs, particularly guest traffic	Assigned challenges: ACLs (traffic filtering policy)
Provide isolated, limited wireless access for after-hours contractors	CR14 change request
Allow controlled communication between segments where legitimately required (e.g., admin to servers)	Appropriate Connectivity + security
Keep the design scalable for future estate growth	Rubric: future -proofing
Keep the design manageable for a small IT function	Non-functional: manageability

7. proposed Network Architecture

The proposed architecture uses a simple two-tier hierarchical design appropriate for a single-site estate of this size: a dedicated edge router providing the Internet connection, Network Address Translation and Internet-facing ACLs; a separate core Layer-3 switch performing inter-VLAN routing (via SVIs) and internal ACL enforcement between segments; and access-layer switches serving each functional area.

This design commits to two physically separate devices rather than a single router-on-a-stick, because a dedicated core switch scales better as VLANs are added, keeps WAN/NAT concerns cleanly separated from internal routing, and matches standard Cisco campus-network practice. This decision is final for Phase 1 and is reflected consistently in both the physical topology (Figure 1) and the logical topology (Figure 2).

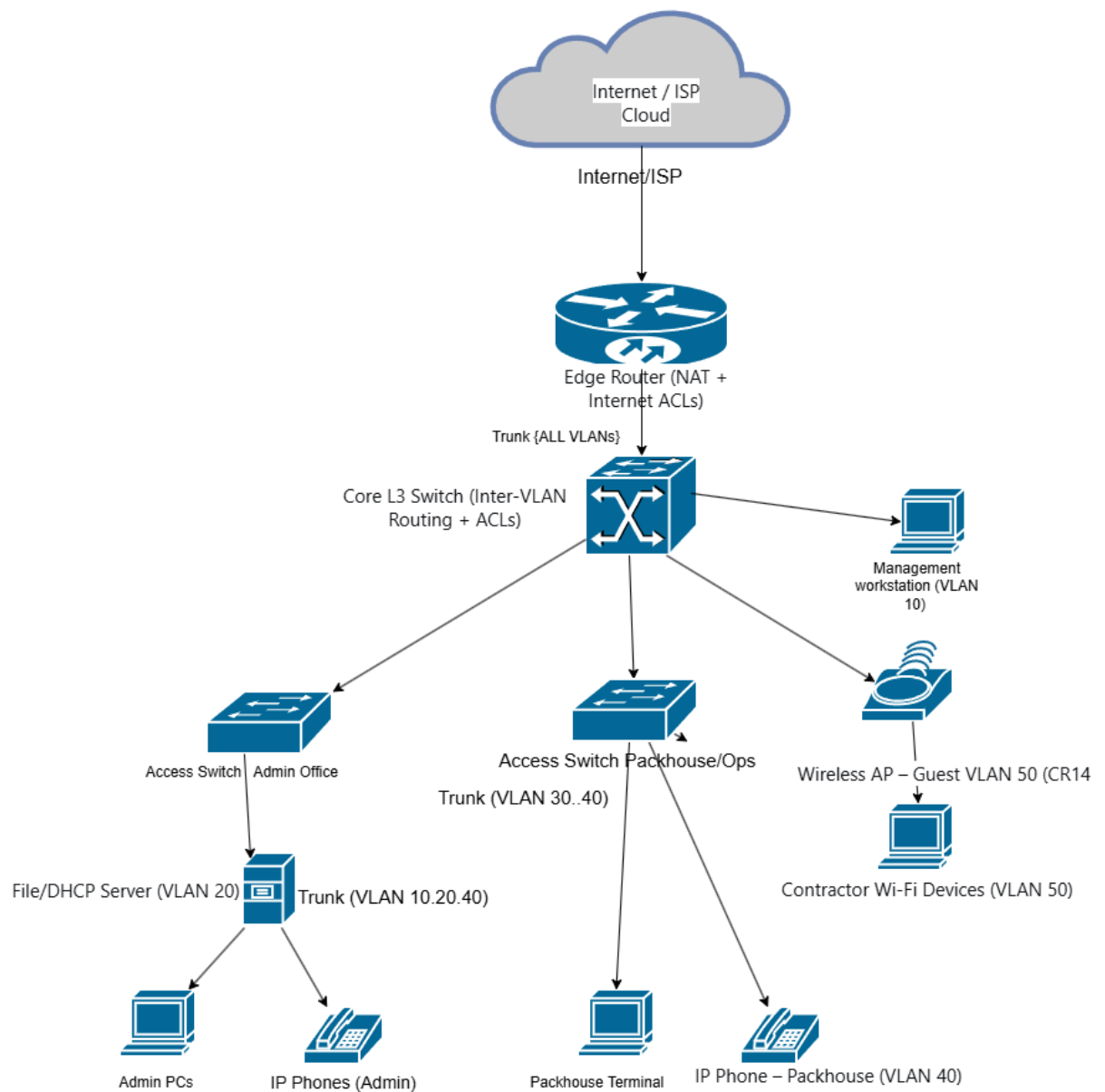
component	Role
Edge Router	Internet/WAN connection, NAT, Internet-facing ACLs
Core Layer – 3 Switch	Inter-VLAN routing (SVIs), internal ACL enforcement, trunk aggregation
Access Switch -Admin/Office	Connects admin PCs, IP phones and the file/DHCP server; VLANs 10, 20, 40
Access Switch – Packhouse /Operations	Connects packhouse terminals and IP phones; VLANs 30, 40
Wireless Access Point -Guest	Serves after-hours contractor devices on the isolated Guest VLAN 50
File /DHCP Server	Provides DHCP leases and file sharing to the Administration VLAN
IP Phones	VoIP endpoints on the dedicated Voice VLAN, distributed across buildings

Devices were limited to what is functionally justified by the client's requirements, no additional redundant links, dynamics routing protocols, or extra serves were introduced , since the brief does not require them and the rubric rewards an appropriate design over an artificially complex one.

8. Physical Topology

As shown in figure 1 below, the physical topology comprises the actual and cabling proposed for the estate: s dedicated edge router connecting to the Internet, a separate core Layer-3 switch, two access switches (Admin/Office and packhouse/Operations), a wireless access point for guest/contractor access, and the end devices (PCs, IP Phones, server, management workstation, contractor WIFI-devices).

MOLEMA CITRUS (RUSTENBURG) – PHASE 1 Physical Topology



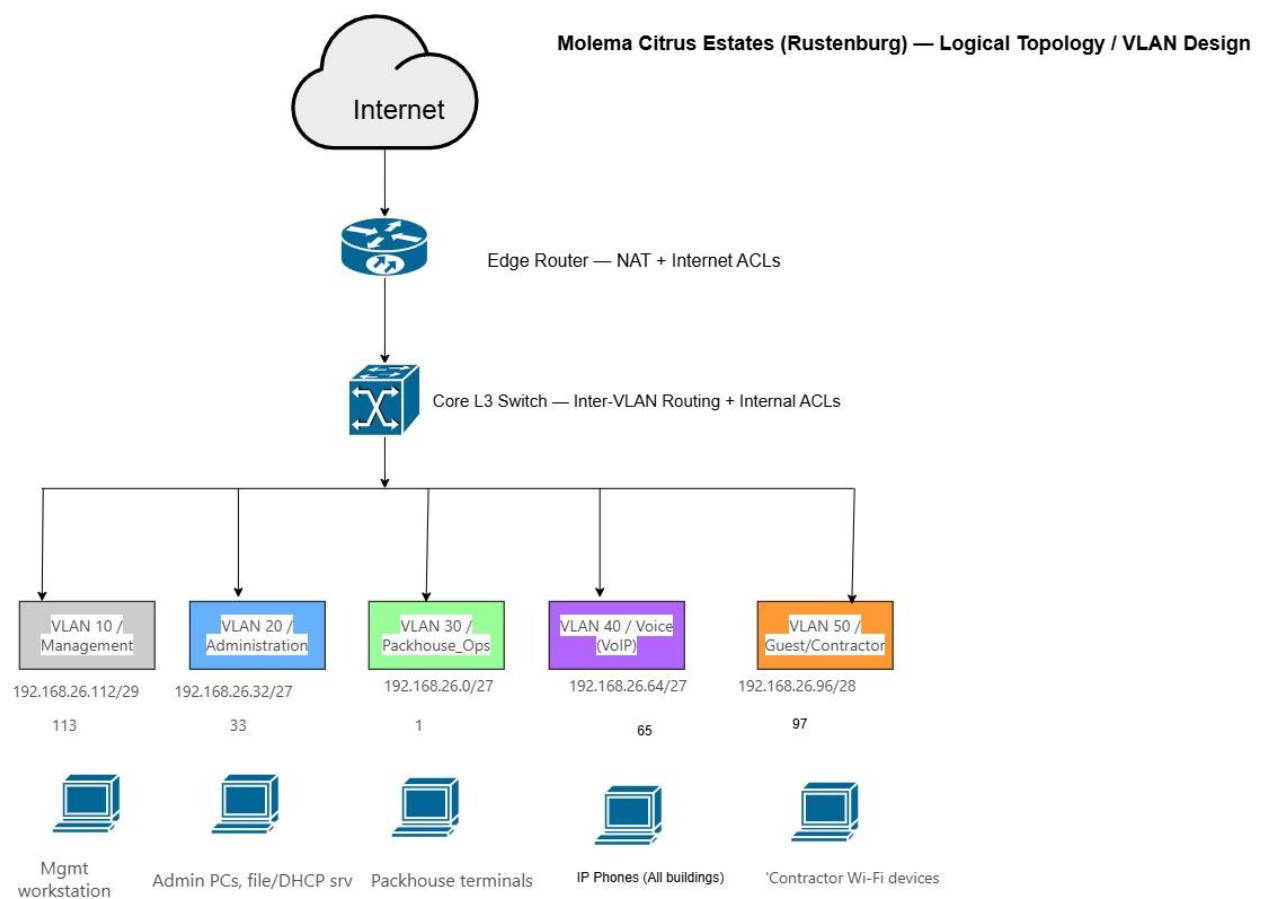
8.1 Topology justification

- Two access switches are used (Admin/Office and Packhouse/Operations) because these are the two functional areas identified in Section 5.2 — one switch per area keeps cabling local to each building and keeps the design proportionate to the estate's size, rather than a single flat switch serving everyone.
- A dedicated wireless access point is used for guest/contractor access rather than extending an internal SSID, so that CR14's 'limited wireless access' can be enforced as its own access point and VLAN, cleanly separated from internal switches.
- A core Layer-3 device is placed centrally so that every access switch and the AP need only a single trunk (or access point) back to the core, keeping the cabling star-like and simple to troubleshoot — appropriate for a single-site estate.
- The file/DHCP server and management workstation connect at the core/admin side because they are estate-wide/administrative services, not tied to a single functional area.
- This physical layout is directly implementable in Cisco Packet Tracer with standard router, multilayer switch, switch, access point and end-device icons.

9. Logical Topology

As shown in figure 2 below, traffic is segmented into VLANs, each mapped to it's own subnet within 192.168.26.0/24, with inter-VLAN routing performed at the core L3 switch and ACL enforcement applied at both the edge router (inter-facing) and the core switch (internal segment to -segment)

MOLEMA CITRUS (RUSTENBURG) – PHASE 1 LOGICAL TOPOLOGY/VLAN DESIGN



9.1 VLAN PLAN

VLAN ID	VLAN NAME	PURPOSE	ASSOCIATED SUBNET
10	Management	Network device administration	192.168.26.112/29
20	Administration	Office/admin staff, file & DHCP server	192.168.26.32/27
30	PACKHOUSE	Packhouse/operations terminals	192.168.26.0/27
40	VOICE	IP phones (all buildings) VoIP separation constraint	192.168.26.64/27
50	GUEST	After-hours contractor Wi-Fi (CR14)	192.168.26.96/28

9.2 Logical design justification

- VLAN 40 (Voice) exists specifically to satisfy the confirmed constraint that VoIP traffic must be separated from data traffic — every IP phone, regardless of which building it is in, is placed on this VLAN and subnet, separate from the PCs/terminals in the same building.
- VLAN 50 (Guest) exists specifically to satisfy CR14 — it is logically and physically isolated (own AP, own subnet, own ACL rules) from every internal VLAN, and is intended for Internet access only.
- VLAN 10 (Management) is separated from user traffic so that switch/router administrative access is not reachable from ordinary user or guest devices, supporting the manageability and security non-functional requirements.
- VLANs 20 and 30 separate administration from packhouse/operations traffic so that broadcast domains stay small and department-level ACL policy (e.g., restricting packhouse terminals from admin file shares, if later required) is possible without redesigning the network.

10.IP Addressing Plan (VLSM)

The client allocated 192.168.26.0/24 (256 addresses).VLSM is used so each segment receives only the block size it needs ,rather than splitting the/24 into equal subnets, which would waste address space. Segments are allocated largest-host-count first

10.1 Host requirements used for sizing

Host counts are the design assumptions from Section 5.2, used only to size subnets:

- Packhouse/Operations ≈ 30 devices (largest segment)
- Administration ≈ 20 devices
- Voice (VoIP) ≈ 15 IP phones
- Guest/Contractor Wi-Fi ≈ 10 concurrent devices
- Management ≈ 5 devices
- Router-to-ISP WAN link = 2 addresses (point-to-point)

10.2 VLSM allocation table

Network/VLAN	Req.Ho sts	Network Address	prefix	Subnet Mask	Usable Host Range	Broadcast	Default Gateway
Packhouse (VLAN 30)	30	192.168.26.0	/27	255.255.255. 224	1 – .30	192.168.26.3 1	192.168.26.1
Administration (VLAN 20)	20	192.168.26.3 2	/27	255.255.255. 224	.33 – .62	192.168.26.6 3	192.168.26.33
Voice (VLAN 40)	15	192.168.26.6 4	/27	255.255.255. 224	65 – .94	192.168.26.9 5	192.168.26.65
Guest (VLAN 50)	10	192.168.26.9 6	/28	255.255.255. 240	.97 – .110	192.168.26.1 11	192.168.26.97
Management (VLAN 10)	5	192.168.26.1 12	/29	255.255.255. 248	.113 – .118	192.168.26.1 19	192.168.26.11 3
Router-to-ISP WAN link	2	192.168.26.1 20	/30	255.255.255. 252	.121 – .122	192.168.26.1 23	n/a (point-to-point)

10.3 VLSM calculation walkthrough

- Packhouse/Operations needs 30 usable addresses. The smallest block satisfying $2^n - 2 \geq 30$ is $n = 5$ ($2^5 - 2 = 30$), i.e. a /27 (32 addresses). Allocated first, from 192.168.26.0, giving 192.168.26.0/27 (.0–.31).
- Administration needs 20 usable addresses. A /28 ($2^4 - 2 = 14$) is too small since $14 < 20$, so the next block, /27 ($2^5 - 2 = 30$ usable), is used: 192.168.26.32/27 (.32–.63), immediately after the first block.
- Voice needs 15 usable addresses. A /28 ($2^4 - 2 = 14$) is one short since $14 < 15$, so a /27 ($2^5 - 2 = 30$ usable) is used again: 192.168.26.64/27 (.64–.95).
- Guest needs 10 usable addresses. A /28 ($2^4 - 2 = 14$) fits since $14 \geq 10$, with 4 addresses of spare capacity: 192.168.26.96/28 (.96–.111).
- Management needs 5 usable addresses. A /29 ($2^3 - 2 = 6$) fits since $6 \geq 5$: 192.168.26.112/29 (.112–.119).
- The Router-to-ISP WAN link needs only 2 addresses (point to point). A /30 ($2^2 - 2 = 2$) fits exactly: 192.168.26.120/30 (.120–.123).
- Remaining space, 192.168.26.124 – 192.168.26.255 (132 addresses), is left unallocated and reserved for future growth (Section 17 of the design justification).

Subnetting calculation summary table

Network	Required Hosts	Formula ($2^n - 2 \geq \text{Required}$)	n	Prefix	Block Size	Starting Address
Packhouse	30	$2^5 - 2 = 30 \geq 30$	5	/27	32	192.168.26.0
Administration	20	$2^4 - 2 = 14 < 20$, so $2^5 - 2 = 30 \geq 20$	5	/27	32	192.168.26.32
Voice	15	$2^4 - 2 = 14 < 15$, so $2^5 - 2 = 30 \geq 15$	5	/27	32	192.168.26.64
Guest	10	$2^4 - 2 = 14 \geq 10$	4	/28	16	192.168.26.96
Management	5	$2^3 - 2 = 6 \geq 5$	3	/29	8	192.168.26.112
WAN Link	2	$2^2 - 2 = 2 \geq 2$	2	/30	4	192.168.26.120

10.3 Default gateway convention

Default gateways are consistently assigned as the first usable address in each subnet (network address + 1) across every VLAN in this design. This convention simplifies administrative documentation, avoids ambiguity during Phase 2 configuration, and aligns with standard Cisco SVI configuration practice.

11.Design and Justification

11.1Hierarchical topology (router + core L3 switch + access switches)

A two-tier hierarchical topology was selected over a flat single switch design

Aspect	Explanation
Reason	The client has multiple distinct traffic types (data, voice, guest, management) that need routing and policy between them.
Client benefit	Keeps the network organized by function, making faults easier to isolate and staff training simpler.
Technical benefit	Concentrates routing/ACL logic at one point (the core), rather than spreading it across many devices.
Alternative considered	A single flat switch with no VLANs
Why this is preferable	A flat design cannot meet the confirmed VoIP-separation constraint or CR14's guest isolation requirement

11.2Five-VLAN segmentation

Traffic is segmented into five VLANs : Management, Administration, Packhouse/Operations, Voice, and Guest.

Aspect	Explanation
Reason	Each VLAN corresponds to a distinct traffic type or security zone identified from the confirmed requirements and reasonable assumptions.
Client benefit	Contractors, phones, and business systems don't interfere with or expose each other.
Technical benefit	Smaller broadcast domains, and a policy boundary (ACL) between every VLAN pair that needs one.
Alternative considered	Two VLANs only (data + voice), leaving guest and management on the data VLAN.
Why this is preferable	Two VLANs cannot satisfy CR14's isolation requirement or the objective of manageability for restricting device management access.

11.3 VLSM addressing instead of fixed size subnetting

192.168.26.0/24 is divided using VLSM into /27, /28, /29 and /30 blocks sized to each segment's needs.

Aspect	Explanation
Reason	A flat split into five equal /27 blocks would only allow 30 hosts per segment and waste addresses on small segments like Management and the WAN link
Client benefit	Leaves over half the address block (132 addresses) free for future growth.
Technical benefit	Matches subnet size to actual host requirements, which is the correct VLSM methodology and directly satisfies the rubric's 'efficient IP addressing' criterion.
Alternative considered	Equal-size subnetting (e.g., five /27s).
Why this is preferable	VLSM avoids wasting an oversized block on Management (needs 5, would get 30) or the WAN link (needs 2, would get 30).

11.4 Dedicated Voice VLAN

All IP phones, regardless of building, are placed on VLAN 40 with its own subnet

Aspect	Explanation
Reason	This is a confirmed client design constraint.
Client benefit	Reliable call quality, and phones are not exposed to or from ordinary data devices.
Technical benefit	Voice traffic can be filtered and prioritised independently of data traffic.
Alternative considered	Placing phones on the same VLAN as PCs in each building.
Why this is preferable	That alternative directly violates the stated constraint and mixes sensitive voice signalling with general data traffic.

11.5 Isolated Guest VLAN with ACL policy

Contractor Wi-Fi devices are placed on VLAN 50, served by a dedicated AP, and denied access to all internal VLANs by ACL, with Internet access only.

Aspect	Explanation
Reason	CR14 specifically requires limited (not full) after-hours wireless access for contractors.
Client benefit	Contractors get the Internet access they need without exposing farm/business systems.

Technical benefit	Satisfies the assigned ACL challenge with a clear, defensible traffic-filtering policy.
Alternative considered	Giving contractors access to the existing Administration Wi-Fi/VLAN.
Why this is preferable	That alternative would give contractors full internal reachability, contradicting the brief's explicit 'limited' wording.

11.6 No dynamic routing protocol

Static/default routing and directly connected SVI routes are used instead of a dynamic routing protocol (e.g., OSPF).

Aspect	Explanation
Reason	A single site with one exit point to the Internet and five directly connected subnets does not require dynamic routing.
Client benefit	Simpler configuration means fewer failure points for a client with a small IT function.
Technical benefit	Reduces overhead and configuration/troubleshooting complexity where it provides no real benefit.
Alternative considered	OSPF or another IGP between the router and core switch.
Why this is preferable	Dynamic routing would add complexity without solving any requirement in the brief — the rubric rewards appropriateness, not unnecessary complexity.

12. References

- North-West University, Department of Computer Science and Information Systems. CMPG 325 — Computer Networks: Individual Semester Project Brief, CMPG325-2026-037 (2026).
- North-West University. CMPG 325 Project Handbook (common project guidance, as referenced in the project brief).
- Cisco Systems. Catalyst 9000 / 3850 Series Switches — Software Configuration Guide (VLAN, SVI and ACL configuration reference).
- Cisco Systems. Cisco IOS IP Addressing Services Configuration Guide (VLSM and subnetting reference).
- Lammle, T. CCNA Routing and Switching Study Guide (subnetting and VLAN design reference).

13.GITHUB/LINK

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